

A 32-Point Counterexample to the Additive–XOR Conjecture

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17 August 2026

Status: candidate full counterexample, likely valid,
to the B_n^σ conjecture in arXiv:math/0308091.

1 The conjecture

Write $[N] = \{0, 1, \dots, N - 1\}$ and let $\oplus$ denote bitwise XOR. For a permutation σ of $[2^n]$, Hinrichs and Wenzel define

$$B_n^\sigma = \{(k, l) \in [2^n]^2 : k + l < 2^n, \sigma(k) \oplus \sigma(l) = \sigma(k + l)\}.$$

They observe that $\#B_n^{\text{id}} = 3^n$, verify the upper bound for all permutations when $n \leq 4$, and conjecture that $\#B_n^\sigma \leq 3^n$ for every n and every permutation $[1, \text{p. } 17]$.

better than ours. It would be interesting to find the optimal estimates at least in the cases of the usual orderings.

2. Although we were not able to give general estimates for the cardinalities of the sets A_n^σ , we conjecture that the identical permutation already gives the maximal possible cardinality. A similar and from the combinatorial point of view very natural question is to find good upper bounds for the cardinalities of the sets

$$B_n^\sigma = \{(k, l) \in [2^n]^2 : k + l \in [2^n], \sigma(k) \oplus \sigma(l) = \sigma(k + l)\}.$$

For linear and piecewise linear rearrangements one can obtain that $\#B_n^\sigma \leq 3^n$ and for the identity $\#B_n^{\text{id}} = 3^n$. Again we conjecture that $\#B_n^\sigma \leq 3^n$ holds for any permutation σ . Basically, this is a question about how big the set of pairs (k, l) can be for which σ behaves like a homomorphism between the integers and the Cantor group. We checked this claim for $n \leq 4$ and for all permutations σ of $[2^n]$ with a computer. The running time for the case $n = 4$ on a PC was about four days.

2 Proof intuition

The conjectured bound first fails at the next unchecked dimension. An exact local search at $n = 5$ found a permutation with six more agreements than the identity. Postcomposition by an invertible linear map of the XOR group does not change any agreement and turns the search output into the compact table below. The finite base count is certified by listing all 32 sum-fibre counts. The counterexample then lifts to every larger dimension: retain the permutation on the lowest five bits and leave all higher bits fixed. A carry out of the low block makes the high-bit equality impossible by parity; with no carry, each additional bit has the usual three choices.

3 Counterexample and infinite lift

Define $q : [32] \rightarrow [32]$ by the following table.

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$q(k)$	0	1	2	3	4	11	10	9	8	15	14	13	12	19	18	5

k	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
$q(k)$	16	7	6	25	24	27	26	29	28	31	30	17	22	23	20	21

Theorem 1. *The displayed q is a permutation and*

$$\#B_5^q = 249 > 243 = 3^5.$$

More generally, for every $n \geq 5$, put $m = n - 5$, write uniquely $x = 32b + a$ with $a \in [32]$ and $b \in [2^m]$, and define

$$\sigma_n(32b + a) = 32b + q(a). \quad (1)$$

Then σ_n is a permutation of $[2^n]$ and

$$\#B_n^{\sigma_n} = 249 \cdot 3^{n-5} > 3^n.$$

Thus the conjecture is false in every dimension $n \geq 5$.

Proof. The displayed row of values contains each integer from 0 to 31 exactly once, so q is a permutation. For $s \in [32]$ define the fibre count

$$c_s = \#\{a \in [0, s] : q(a) \oplus q(s - a) = q(s)\}.$$

Direct substitution in the displayed table gives, in blocks of eight,

s	0	1	2	3	4	5	6	7
c_s	1	2	2	4	2	2	4	4
s	8	9	10	11	12	13	14	15
c_s	8	4	6	6	10	2	4	6
s	16	17	18	19	20	21	22	23
c_s	6	12	10	6	6	10	12	8
s	24	25	26	27	28	29	30	31
c_s	8	12	14	10	8	14	14	32

Their sum is

$$\#B_5^q = \sum_{s=0}^{31} c_s = 249.$$

For comparison, under the identity permutation a decomposition $s = a + c$ is good exactly when the binary supports of a and c are disjoint. Each 1-bit of s can be assigned to either summand, so the corresponding fibre has $2^{\text{popcount}(s)}$ elements. Therefore

$$\#B_n^{\text{id}} = \sum_{s < 2^n} 2^{\text{popcount}(s)} = (1 + 2)^n = 3^n.$$

This proves the strict inequality in the base case.

It remains to prove the lifting formula. Let

$$x = 32b + a, \quad y = 32d + c$$

with $a, c \in [32]$ and $b, d \in [2^m]$, and assume $x + y < 2^n$. If $a + c \geq 32$, then comparison of the high blocks in $\sigma_n(x) \oplus \sigma_n(y) = \sigma_n(x + y)$ would require

$$b \oplus d = b + d + 1.$$

The two sides have opposite parity, so no such pair is good.

If $a + c < 32$, the equality separates into

$$q(a) \oplus q(c) = q(a + c), \quad b \oplus d = b + d.$$

The first condition has 249 ordered solutions (a, c) . The second says that b and d have disjoint binary supports. Independently at each of the m high bits, the pair of bits is $(0, 0)$, $(1, 0)$, or $(0, 1)$, giving exactly 3^m solutions. Such a carry-free high sum is automatically below 2^m . Hence

$$\#B_n^{\sigma_n} = 249 \cdot 3^m = 249 \cdot 3^{n-5} > 243 \cdot 3^{n-5} = 3^n,$$

as claimed. □

Verification, novelty, and scope

The accompanying verifier checks that the table is a permutation, reconstructs all 32 fibre counts and their sum using integer XOR, verifies the identity benchmark, and exhaustively checks the lifted counts through $n = 10$. These finite checks are not used for the all- n lifting argument, which is proved above by carry and parity analysis.

The paper states a separate conjecture that the identity maximizes the four-term cardinality $\#A_n^\sigma$. The present q has $\#A_5^q = 6912 < 7776 = \#A_5^{\text{id}}$, so this packet makes no claim about that conjecture.

A bounded novelty search checked the run indexes, the exact conjecture, author/title searches, and the source's citation graph. OpenAlex lists one later citation, Sanders' arXiv:1901.03109, which proves full Walsh–trigonometric nonequivalence by a different additive-combinatorial route but does not state this finite conjecture or the counterexample. No later explicit resolution was located. Novelty is therefore provisional pending expert search.

References

- [1] A. Hinrichs and J. Wenzel, *On the non-equivalence of rearranged Walsh and trigonometric systems in L_p* , *Studia Mathematica* 159 (2003), 435–451; arXiv:math/0308091.